



EUNOIA JUNIOR COLLEGE

JC2 Preliminary Examination 2018

General Certificate of Education Advanced Level

Higher 1

MATHEMATICS

8865/01

Paper 1 [100 marks]

12 September 2018

3 hours

Additional Materials: Answer Paper

List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and question number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

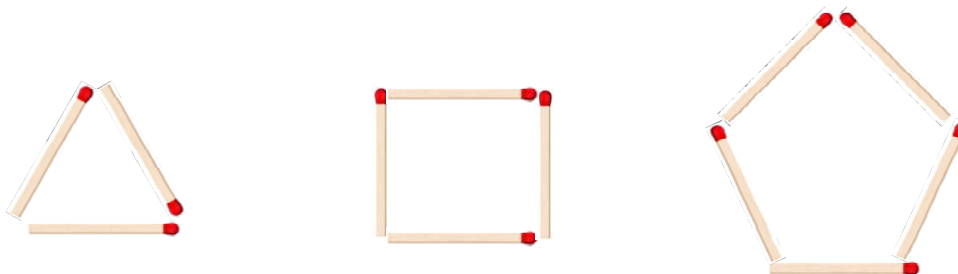
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages and **1** blank page.

Section A: Pure Mathematics [40 marks]

- 1 Show that there are no real values of k for which $(k-8)x + 2k - 2x^2$ is always negative. [4]

- 2 Chia has a total of 100 blue, red and yellow matchsticks. All the matchsticks are indistinguishable apart from their colour. He uses all the blue matchsticks to form triangles, all the red matchsticks to form squares and all the yellow matchsticks to form pentagons as shown in the diagram.



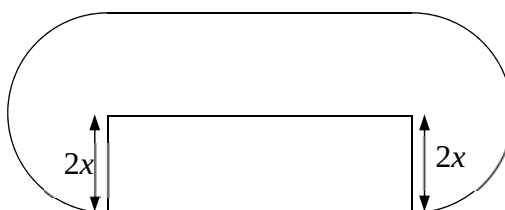
The total number of triangles and pentagons formed is four times the number of squares formed. If

Chia were to exchange $\frac{1}{5}$ of his yellow matchsticks for red matchsticks, he will have the same

number of blue and red matchsticks. Find the number of matchsticks that Chia has for each colour

and deduce the number of triangles, squares and pentagons formed. [5]

- 3 A new flower-bed is being designed for a large garden. The flower-bed will occupy a rectangle flanked by two semi-circles as shown in the diagram. Each semi-circle has radius $2x$ m and the total perimeter of the flower-bed is 40π m.



- (i) Show that the area enclosed by the flower-bed is $4x(10\pi - x)$ m². [3]

The area of the flower-bed is to be as large as possible.

- (ii) Use a non-calculator method to find the exact maximum area of the flower-bed, giving your answer in terms of π . Verify that this is the maximum value. [4]

- 4 The equation of a curve C is $y = 2 - 2e^{-2x+1} + e^{-x}$.
- (i) Sketch the curve C , stating the equations of any asymptotes and the coordinates of any turning points. [3]
 - (ii) Find the equation of the tangent to the curve at the point where $x = 1$, giving your answer in the form of $ax + ey = b + 2e$, where a and b are integers. [4]
 - (iii) Use a non-calculator method to find the exact area of the region bounded by C , the x -axis and the lines $x = 1$ and $x = 3$. [4]
- 5 Mr Tan started a manufacturing company to produce LED lights on 1 January 2010. The annual revenue from the sale of these lights is S (in thousands) per year. Mr Tan believes that the connection between S and the time, t years from 1 January 2010, can be modelled by

$$S = 40 \ln(t+1) - 0.5t^2, \text{ for } 0 \leq t \leq 10.$$

- (i) Sketch the graph of S against t . [2]
- (ii) Using differentiation, find the year in which the company will earn the maximum annual revenue, justifying that this is a maximum. [4]
- (iii) Explain if the model is suitable in the real world context and suggest a reason why the annual revenue would start to decrease after a period of time. [4]

The company hired an economist to suggest a model for the rate, C (in thousands), of the total manufacturing cost to produce LED lights. The economist believes that the connection between C and the time, t years from 1 January 2010, can be modelled by

$$C = 5 + e^{-2t}, \text{ for } 0 \leq t \leq 10.$$

- (iv) Sketch the graph of C against t . [2]
- (v) Find the exact area of the region bounded by the curve, the t -axis and the lines $t = 0$ and $t = 4$.
Give an interpretation of the area that you found, in the context of the question. [3]

Section B: Probability and Statistics [60 marks]

- 6 The Body Mass Index (BMI) is a measure of body fat based on height and weight that applies to men and women. In a certain country, a person with a BMI of 25 and above is deemed to be overweight. Anyone with a BMI above 30 is considered obese. The Health Promotion Board recorded the BMI of a large number of adult males. The results show that 36.2% of them are overweight and 5.5% of them are obese. Assuming that the BMI of adult males are normally distributed, find the mean and variance of the distribution. [4]

- 7 (i) Find the number of different 6 letter code that can be formed using the word PASSES if repetition of the letters are allowed. [1]

If no letters are repeated, find the number of 6 letter code that can be formed from the same word if

- (ii) all the 3 S's are consecutive, [2]
 (iii) the first letter is S and the last letter is E, [2]
 (iv) the letters P, A and E are separated. [2]

- 8 A mathematics teacher at certain Junior College would like to examine the relationship between the number of optional homework problems students do during the semester, x , and their final examination marks, y . She randomly selects 10 students for study and asks them to keep track of the number of these problems completed during the course of the semester. The data is shown in the following table

x	51	55	58	62	68	76	78	84	85	91
y	62	64	66	66	67	72	72	73	76	75

- (i) Sketch a scatter diagram to illustrate the above data and calculate the product moment correlation coefficient between x and y . [3]
- (ii) Find the equation of the regression line of y on x , in the form $y = a + bx$. Sketch this line on your scatter diagram. [2]
- (iii) Use a suitable regression line to estimate the final examination marks of a students who missed the examination but has completed 70 optional homework problems. Comment on the reliability of this estimate. [2]
- (iv) Explain the meaning of a in the context of this question. [1]

- 9** A university qualification involves two separate examinations, namely CAT I and CAT II. To be a successful, a student must first pass CAT I, and after passing CAT I, the student must then pass CAT II. Students who fail CAT I at the first attempt always make a second attempt. Students are allowed at most two attempts at CAT I but only one attempt at CAT II. The probability that a student passes CAT I, on either attempt, is p .

The probability that the student passes CAT II is

- p^2 if the student passes in CAT I at the first attempt.
- 0.45 if the student passes in CAT I at the second attempt.

- (i) Draw a tree diagram to represent this information. [3]
- (ii) Find the value of p if the probability that a randomly selected student is successful in the university qualification is 0.55. [2]
- (iii) Find the probability that a randomly selected student is successful in the university qualification, given that the student failed CAT I in the first attempt. [3]

- 10** A shop sells balloons in packs of 50 and, on average, 4% of the balloons are faulty.

- (i) State, in context, two assumptions needed for the number of faulty balloons in a pack to be well modelled by a binomial distribution. [2]

Assume that the number of faulty balloons in a pack follows a binomial distribution.

- (ii) Find the expected number of balloons that are faulty. [1]
- (iii) Find the probability that a pack of 50 balloons contains more than 2 faulty balloons. [2]

The balloons are sold in cartons and each carton contains 20 packs. In a randomly selected carton,

- (iv) find the probability that less than 25% of the packs have more than 2 faulty balloons, [3]
- (v) find the probability of having at least 3 packs but fewer than 10 packs that have more than 2 faulty balloons. [2]

- 11** A telecommunication company provides mobile services for its customers to make calls to City *J* and City *C*. The duration of calls, in minutes, made by its customers to City *J* and City *C* have independent normal distributions. The means and standard deviations of these distributions are shown in the following table.

	Mean	Standard deviation
City <i>J</i>	8	1.5
City <i>C</i>	10	1.8

- (i) Find the probability that a randomly chosen call to City *J* takes more than 11 minutes. [1]
- (ii) Find the probability that three randomly chosen calls to City *J* each takes less than 11 minutes. [2]
- (iii) The probability that the total time for three randomly chosen calls to City *J* is less than 33 minutes is denoted by p . Without calculating its value, explain why p will be greater than your answer to part (ii). [1]
- (iv) Find the probability that the total duration of three randomly chosen calls to City *J* and two randomly selected calls to City *C* is more than 48 minutes. [3]

A customer, Ashton, makes calls to City *J* and City *C* for work purposes. The telecommunication company charges a rate of \$0.20 per minute for every call to City *J* and a rate of \$0.30 per minute for every call to City *C*.

Let J represent the cost of one call from Ashton to City *J*.

Let C represent the cost of one call from Ashton to City *C*.

- (v) Find $P(J_1 + J_2 + J_3 - 2C > 1.50)$ and explain, in the context of this question, what your answer represents. [4]

- 12** An engineer is testing the lifetime of a certain brand of battery that is claimed to have a lifetime of 150 hours. A random sample of 70 batteries of this type is tested and the lifetime, x hours, of each battery is measured. The results are summarised by

$$\sum x = 10\,317, \quad \sum x^2 = 1\,540\,231.$$

- (i)** Write down appropriate hypotheses for the test, defining any symbols you use. [2]
- (ii)** Test, at the 10% significance level, whether the population mean lifetime of the batteries is less than 150 hours. [4]
- (iii)** State the meaning of the p -value in the context of the question. [1]
- (iv)** State, with a reason, whether in using the above test, it is necessary to assume that the lifetime of the batteries has a normal distribution. [1]

A new random sample of 70 batteries is chosen and the mean lifetime of this sample is t hours. The engineer introduces a new testing system and concludes that the population mean lifetime of the batteries has changed at the 10% level of significance. It was known that the population standard deviation of the lifetime of a battery is 15 hours.

Find the set of values of t for the engineer's conclusion to be valid, correct to one decimal place. [4]